PYTHON AND AI · EPISODE 04

Your First Python Program

Getting Started

KEY POINTS

- At the **>>>** prompt, each line runs as soon as you press Enter. `print("Hello World")` shows text on screen.
- Use **straight quotes** and a lowercase **print**. Curly quotes and `Print` both stop with an error.
- Save a program as a **.py** file and run it with `python hello.py` **from the folder the file is in**.
- The terminal finds `python` by searching the folders in **PATH**, top to bottom. The first match wins.

01 Hello, World

▶ VIDEO 00:29

Open a terminal, type `python` and press Enter. The three arrows `>>>` mean Python is listening.

Terminal

```
> python
>>> print("Hello World")
Hello World
>>> print("Hello Python")
Hello Python
>>> exit
```

- `print` shows something on the screen. The **brackets** hold what to show, and the **quotes** mark it as text, so it prints exactly as written.
- `>>>` at the start of a line: you're talking to Python. A folder path like `PS C:\Users\you>`: you're back in the regular terminal.

02 The Wrong Quotes

▶ VIDEO 01:26

WATCH OUT

Curly quotes `“ ”` come from Word, Google Docs or notes apps, which swap straight quotes for curly ones. Python stops: `SyntaxError: invalid character '“' (U+201C)`. Type code in the terminal or a plain text editor, never in a word processor.

Python is **case sensitive**: `Print("Hello World")` gives
`NameError: name 'Print' is not defined. Did you mean: 'print'?`

03 Save It in a File

▶ VIDEO 02:17

At the prompt, each line runs once and is gone when you close the window. A real program lives in a file that ends in **.py**: plain text you can run again and again. Create a `python` folder on the desktop, then type these three lines in Notepad and save them there as `hello.py`:

hello.py — PYTHON

```python
# Episode 04 · Your first Python file: three print lines, run with python hello.py
print("####################")
print("### Hello Python ###")
print("####################")
```

04 Run Your File

▶ VIDEO 02:56

Terminal

```
> python hello.py
... can't open file 'C:\\Users\\you\\hello.py': [Errno 2] No such file or directory
> cd Desktop\python
> python hello.py
####################
### Hello Python ###
####################
```

KEY IDEA

The terminal always works inside one folder, and the prompt shows which one. Run your file from the folder it's in: `cd` (change directory) there first, or right-click inside the folder in File Explorer and choose **Open in Terminal**. On a Mac, type `python3 hello.py`.

05 The Hidden .txt

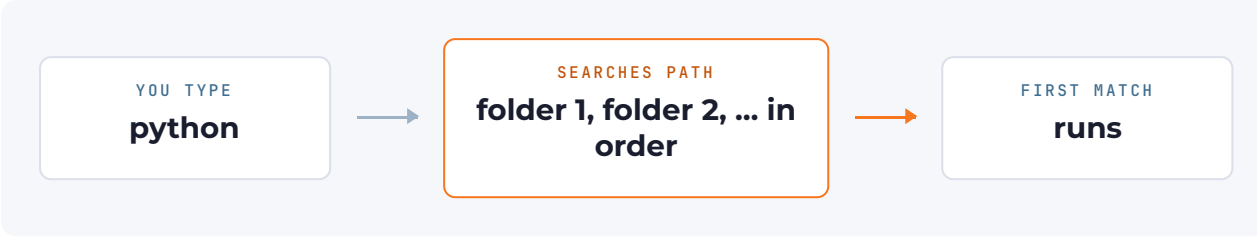
▶ VIDEO 03:56

A file made with right-click › New › Text Document and renamed to `hello.py` is really `hello.py.txt`, because Windows hides file endings by default. In File Explorer, click **View › Show › File name extensions**, then rename the file to `hello.py`.

06 Finding Python

▶ VIDEO 04:31

FIGURE 4.1 **How the terminal finds a command**



No folder has it? "The term is not recognized." The Python install manager sets PATH up for you.

An older Python earlier in the list can win instead. Check with `python --version`. If it isn't the version you expect, open **Installed apps** and remove the old Python.

CODE 3 files on GitHub github.com/abitclearer/python-and-ai/tree/main/04-your-first-python-program